

Comparative Study of Transfer Learning Architectures for 120-Class Handwritten Javanese Script Recognition Under Severe Class Imbalance

1st Fathan Fardian Sanum

School of Computing
Telkom University
Bandung, Indonesia
fathansanum@student.
telkomuniversity.ac.id

2nd Syahdan Rizqi Ruhendy

School of Computing
Telkom University
Bandung, Indonesia
syahdanruhendy@student.
telkomuniversity.ac.id

3rd Ardia Alif Ramadhan

School of Computing
Telkom University
Bandung, Indonesia
ardiar Ramadhan@student.
telkomuniversity.ac.id

Abstract—Javanese script (*Hanacaraka*) is a traditional writing system facing the threat of functional extinction due to a declining number of generations able to read and write it, making digitization through automated character recognition an urgent preservation strategy. Three research gaps not yet addressed simultaneously in the literature are full 120-class recognition under natural class imbalance, with an imbalance ratio of $33.10\times$ (largest class 695 images versus smallest class 21 images) between the Vowel A classes and the five other vowel groups, structurally safe augmentation that preserves Javanese script characteristics, and comparative evaluation across transfer learning architectures. This study proposes a four-step structure-preserving data preparation pipeline that explicitly avoids geometric augmentation to maintain the visual integrity of the script, combined with a weighted loss function to compensate for class imbalance. Four transfer learning architectures (EfficientNet-B0, MobileNetV3-Large, ResNet-18, and VGG-16) are evaluated comparatively using an identical pipeline, with metrics reported separately for majority and minority classes. EfficientNet-B0 achieves 99.06% accuracy with a macro F1-score of 0.9823, surpassing prior studies using artificially balanced datasets, while simultaneously maintaining an F1 gap between majority and minority classes of only 0.0291, the best generalization balance among all architectures tested. These results demonstrate that extreme class imbalance can be overcome with a targeted pipeline, while per-vowel-group analysis shows that the E-taling group is consistently the hardest to recognize across all four architectures, a pattern hidden by aggregate metrics.

Index Terms—*Abugida*, augmented character recognition, class-imbalanced dataset, EfficientNet-B0, handwritten character recognition, Javanese script, MobileNetV3-Large, ResNet-18, structure-preserving augmentation, *transfer learning*, VGG-16, *weighted loss*

I. INTRODUCTION

Javanese script (*Hanacaraka*) is an *abugida* writing system combining 20 base consonants with six vowel markers into 120 syllabic classes [1]. The dominance of Latin script in modern education has caused a cross-generational decline in proficiency [2], [3], making digitization via *Optical Character*

Recognition (OCR) an urgent preservation intervention [4]. The primary challenge is that the only public dataset covering all 120 classes, *Indonesian Local Script Characters* [3], exhibits extreme imbalance, where Vowel A averages 500 images per class while 100 non-A classes have only around 21 images each, yielding an imbalance ratio of $33.10\times$.

Transfer learning based on CNNs has proven effective for handwritten regional script recognition with limited data [5], [6], significantly outperforming *shallow learning* methods. Combined with a weighted loss function, this approach provides a suitable framework for simultaneously addressing the visual complexity of Javanese script and its class imbalance.

Prior studies leave three gaps not yet addressed simultaneously. Susanto et al. [8] achieved 99.65% accuracy on 120 classes but on an artificially balanced dataset, while Wisesty et al. [3] found that geometric augmentation actually degrades Javanese script recognition performance. These three gaps are 120-class recognition under natural class imbalance, structurally safe augmentation strategies, and comparative cross-architecture evaluation.

This study closes all three gaps by proposing a four-step structure-preserving data preparation pipeline and evaluating four architectures (EfficientNet-B0, MobileNetV3-Large, ResNet-18, VGG-16) comparatively. EfficientNet-B0 achieves 99.06% accuracy together with the smallest F1 gap across vowel groups of only 0.0291, while per-vowel-group analysis reveals that the E-taling group is the largest source of error across all architectures.

II. RELATED WORK

The following studies are the most closely related to this work because they directly target the full 120 syllabic class coverage of Javanese script, serving as the primary benchmarks to surpass. Susanto et al. [9] developed a custom CNN with four convolutional layers and two *fully connected* layers

for 120-class classification, achieving 97.29% accuracy on a balanced dataset of 14,400 images from 30 writers. The same group [8] subsequently proposed a 12-layer CNN with selective augmentation, where rotation was restricted to $\pm 3^\circ$ to avoid altering the visual meaning of the script, achieving 99.65% accuracy. Despite the high accuracy figures, all experiments were conducted on datasets artificially balanced with an equal number of samples per class, so model performance under naturally imbalanced data distributions has never been explored.

Beyond studies specifically targeting 120 classes, several studies covering the broader Nusantara script landscape provide valuable insights into architecture selection, augmentation behavior, and model capability limits on 20-class Javanese script. Wisesty et al. [3] proposed MNetNCR based on MobileNetV3 [10] for five Nusantara scripts and achieved $F1=0.9788$ for Javanese script without any augmentation. More importantly, applying augmentation to this model actually degraded performance to 0.9579 because geometric transformations destroyed the visual characteristics distinguishing between classes, serving as the primary empirical foundation for the augmentation strategy in this study. Nindya et al. [6] applied VGG-16 [11] to multi-script classification of Balinese, Javanese, and Sundanese scripts, achieving 93.19% accuracy for the Javanese subset, and introduced the *Character Likeness Rate* (CLR) metric identifying the “ha”/“la” pair (CLR 25.3%) and “sa”/“da” (CLR 15.8%) as the most confusion-prone characters due to visual similarity. Sulistiyo et al. [7] compared various *shallow learning* methods on the Javanese script dataset and found that even the best-performing XGBoost model struggled significantly with visually similar characters such as “tha” (F1-score 13.64%) and “nga” (F1-score 29.63%), demonstrating the inherent limitations of feature-based approaches for fine-grained character discrimination. Fateah et al. [5] demonstrated that ResNet-18 [12] fine-tuned with a *transfer learning* strategy can achieve 91% accuracy and 91% F1-score from only 1,562 images of Javanese script, making it the sole ResNet-based *transfer learning* reference explicitly targeting Javanese script recognition prior to this study. All studies in this section, while informative, are limited to balanced datasets and have never tested model robustness under extreme class imbalance.

Class imbalance is a critical aspect overlooked by almost all Javanese script studies, yet it is the natural characteristic of the publicly available dataset. Faizin et al. [2] is the only study to explicitly address this issue, developing SkelBAGAN (*Skeleton-based Balancing GAN*) for a Javanese script dataset with imbalance ratios up to $81.78\times$. SkelBAGAN integrates a character skeleton-based *autoencoder* and *adversarial loss* to generate synthetic images preserving script structure integrity, and demonstrably improves recognition performance on minority classes. Its limitation lies in the GAN architecture complexity requiring significant computational resources, and its scope remaining limited to 20 base classes without testing on the full 120-class syllabic scenario. This study fills that gap with a lighter approach, namely structure-preserving

TABLE I
CLASS DISTRIBUTION OF DATASET BY VOWEL GROUP

Vowel Group	Classes	Images/Class	Total
Vowel A	20	426–695	9,995
Vowel E	20	21–22	440
Vowel I	20	21–22	440
Vowel O	20	21–22	440
Vowel U	20	21–22	440
Vowel E-taling	20	21–22	440
Total	120	21–695	12,191

IR = $33.10\times$, median = 22.

augmentation and a weighted loss function, without requiring additional generative network training.

To understand the positioning of this study in the broader landscape, a review of OCR studies across Southeast Asian abugida scripts provides context on how far research in this area has advanced. Wicaksono et al. [1] conducted a systematic literature review on six Southeast Asian abugida scripts and found no studies in the past five years addressing word segmentation for either Javanese or Sundanese script, confirming that isolated character recognition remains the frontier of Javanese OCR research. Widiarti et al. [4] established a baseline evaluation for printed (not handwritten) Javanese OCR, noting that handwriting style variation adds a far more complex layer of challenges compared to printed script. Together, these two studies underscore that the domain of handwritten Javanese character recognition, particularly at 120-class coverage with imbalanced data, remains very open and serves as the direct motivation for this study.

III. DATASET AND EXPERIMENTAL SETUP

This study uses the *Indonesian Local Script Characters* dataset, curated by Aditya Firman Ihsan et al. at Telkom University and available on Mendeley Data, which was also used by prior studies [3]. For the Javanese script subset, the dataset consists of two parts with a total of 13,257 images, comprising the `all_class/` subdirectory containing 12,191 images across 120 syllabic classes (6 vowels $\times$ 20 consonants), and the `variations/` subdirectory containing 1,066 handwriting style variation images for 8 consonants (ba, ga, ha, ka, la, nya, ra, ta) in the Vowel A group. Images predominantly use JPEG format (91%) with RGB color channels (98.3%) and an original average resolution of 362×347 pixels, subsequently standardized to 224×224 pixels as model input.

The class distribution in this dataset is severely imbalanced by nature, as summarized in Table I. The twenty Vowel A classes have 426–695 images per class ($\mu = 499.8$), while 100 non-A classes have only 21–22 images per class, yielding an imbalance ratio of $33.10\times$ (largest/smallest class = $695/21$) with a per-class median of only 22, far below the mean of 101.59. This imbalance is the primary challenge that all pipeline stages are designed to address.

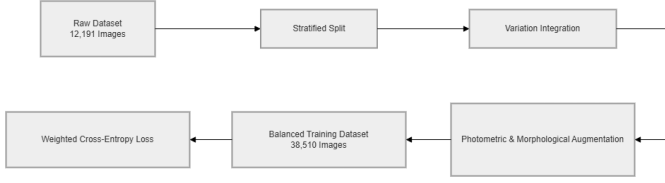


Fig. 1. Four-step data preparation pipeline applied sequentially before model training.

The dataset was split per-class using stratified sampling at a 70:15:15 ratio into training (8,483 original images), validation (1,789 images), and test (1,919 images) sets. Per-class stratification ensures all 120 classes are represented in every subset, including minority classes with only 21 images. After augmentation on the training set, the total training samples increased to 38,510 images. All images were converted to grayscale and replicated across three channels to maintain compatibility with ImageNet-pretrained backbones, then normalized using ImageNet statistics ($\mu = [0.485; 0.456; 0.406]$, $\sigma = [0.229; 0.224; 0.225]$).

IV. PROPOSED METHOD

A. Pipeline Overview

The primary challenge in this study lies not in the model architecture, but in the data conditions, specifically the extreme class imbalance between Vowel A and non-A classes that causes conventional training strategies to produce models systematically biased toward majority classes. To address this, the study proposes a four-step data preparation pipeline executed sequentially before the training process begins. Each step is designed to handle one specific aspect of the data distribution problem, namely maintaining minority class representation in the split, adding data diversity without new collection, filling minority classes with structurally safe augmentation, and applying proportional penalty weights during training. Fig. 1 illustrates the overall flow of this pipeline.

B. Step 1: Per-Class Stratified Split

The dataset is split into training (70%), validation (15%), and test (15%) sets using per-class stratified sampling. This ensures all 120 classes are represented in every subset, even for minority classes with only 21 images, yielding ≈ 15 training images, 3 validation images, and 3 test images per minority class before augmentation. This split yields 1,789 validation images and 1,919 test images.

C. Step 2: Variation Data Integration

A total of 1,066 images from `variations/` are added to the training set for the 8 corresponding Vowel A consonant classes (ba, ga, ha, ka, la, nya, ra, ta). This step increases handwriting style diversity for those classes without requiring new data collection.

TABLE II
STRUCTURE-PRESERVING AUGMENTATION STRATEGY

Category	Transformation	Status
Photometric	Brightness jitter ($\pm 30\%$)	✓
	Contrast jitter ($\pm 30\%$)	✓
	Gaussian blur (radius 0.5–1.5 px)	✓
	Gaussian noise ($\sigma = 1\text{--}5\%$)	✓
	Sharpening ($1.2\times\text{--}2.0\times$)	✓
Morphological	Dilation (MaxFilter $3\times 3 / 5\times 5$)	✓
	Erosion (MinFilter $3\times 3 / 5\times 5$)	✓
Spatial	Random crop + resize (max 10% margin)	✓
Geometric (prohibited)	Rotation ($> \pm 3^\circ$)	×
	Horizontal / vertical flip	×
	Shear / affine transform	×
	Elastic distortion	×

D. Step 3: Structure-Preserving Augmentation

This is the primary methodological contribution of the study. Augmentation is applied *exclusively to the 100 non-A classes* until each class reaches a target of $T = 300$ training images. New images are created by repeatedly sampling from existing training images and applying transformations until each class reaches T samples. The value $T = 300$ was selected to yield approximately a $14\times$ increase over the original ≈ 21 minority images per class, representing a deliberate balance between expanding data diversity and limiting near-duplicate samples generated from a small source pool.

The most critical design constraint is the prohibition of geometric transformations in any form. Wisesty et al. [3] empirically demonstrated that augmentation actually degrades Javanese script recognition because transformations alter visual characteristics that distinguish between classes. This is reinforced by the finding of Nindya et al. [6] that the “ha” and “la” pair are nearly identical horizontal reflections, so applying *horizontal flip* would directly create cross-class ambiguous labels.

Table II lists the permitted and prohibited transformations. Permitted transformations target *appearance variability* (ink density, scan noise, pen stroke thickness) without altering the spatial layout of the character. Each augmented image is created by applying 1–3 transformations selected randomly from the permitted list. Augmentation is performed *offline* (saved to disk) to ensure reproducibility. The final training set after augmentation totals 38,510 images, an increase of approximately $4.0\times$ from 9,549 images before augmentation (8,483 original training images plus 1,066 variation images).

E. Step 4: Weighted Loss Function

After augmentation, class weights are computed using:

$$w_c = \frac{N}{C \cdot n_c} \quad (1)$$

TABLE III
CNN ARCHITECTURES COMPARED

Architecture	Parameters	Literature Relevance
EfficientNet-B0 [13]	~5.3M	–
MobileNetV3-Large [10]	~5.4M	Wisesty et al. [3]
ResNet-18 [12]	~11.7M	Fateah et al. [5]
VGG-16 [11]	~138M	Nindya et al. [6]

where N is the total training samples, $C = 120$ is the number of classes, and n_c is the number of training samples for class c . This formula is equivalent to `compute_class_weight('balanced')` in scikit-learn, which assigns greater loss penalties to misclassification of minority classes. These weights are applied as a parameter to the `CrossEntropyLoss` function during training.

F. Model Architectures

Once the data pipeline is established, the next stage is selecting the model architectures to evaluate. Four CNN architectures with ImageNet-1K pretrained weights are selected based on two complementary criteria. The first criterion is representation of conceptually distinct architecture families, including compound scaling efficiency (EfficientNet), residual connections (ResNet), *depthwise separable convolution* (MobileNetV3), and deep convolutional blocks (VGG). The second criterion is direct relevance to prior Javanese script research, enabling fair comparison against architectures already reported in the literature. The smallest variant of each family was deliberately chosen so that all models could be trained on a single 8 GB GPU at the same input resolution, allowing performance differences to be attributed to architectural character rather than computational budget. EfficientNet-B0 was selected as a representative of compound scaling that balances depth, width, and resolution at a small parameter count. ResNet-18 was chosen because it is the backbone reported by Fateah et al. [5] for Javanese script, whereas MobileNetV3-Large follows the choice of Wisesty et al. [3] and VGG-16 follows Nindya et al. [6], so all three enable direct comparison with already published results. Table III summarizes the four architectures and their literature relevance. Each architecture’s classification head is replaced with a linear layer mapping to 120 classes, while all backbone weights are initialized from ImageNet-1K pretraining.

G. Training Strategy

All models follow a uniform two-phase fine-tuning strategy to ensure fair cross-architecture comparison. The split into two phases is designed to prevent the large gradients from the still-random classification head from corrupting the already informative pretrained weights. In Phase 1 over 10 epochs, all backbone parameters are frozen and only the classification head is trained with a higher learning rate. This phase aims to build a stable head initialization before gradients propagate

TABLE IV
HYPERPARAMETER CONFIGURATION PER ARCHITECTURE

Architecture	Opt.	LR Phase 1	LR Phase 2	WD	ES
EfficientNet-B0	AdamW	10^{-3}	10^{-4}	10^{-4}	20
MobileNetV3-Large	AdamW	10^{-3}	8×10^{-5}	10^{-5}	20
ResNet-18	SGD	2×10^{-2}	10^{-3}	5×10^{-4}	20
VGG-16	SGD	10^{-3}	10^{-4}	5×10^{-4}	20

Opt. = optimizer. WD = weight decay. ES = early stopping patience.

Label smoothing 0.1 and batch 32 on all models, except

VGG-16 uses batch 16, dropout 0.5, and grad clipping 2.0.

to the more sensitive pretrained feature layers, and for the AdamW-based models a three-epoch *warmup* is also used so that the learning-rate transition proceeds smoothly. In Phase 2 running up to 90 epochs, all layers are unfrozen for full fine-tuning with a lower learning rate. The optimizer is matched to each architecture, namely AdamW for EfficientNet-B0 and MobileNetV3-Large which are more stable under adaptive learning rates, and momentum SGD for ResNet-18 and VGG-16 which yields better generalization on those two networks. During Phase 2, a ReduceLROnPlateau scheduler with factor 0.5 adaptively reduces the learning rate when validation loss stagnates, while *early stopping* with *patience* 20 halts training if no improvement occurs, and *label smoothing* of 0.1 is applied to all models to curb overconfidence toward majority classes. All models are trained using the weighted `CrossEntropyLoss` per (1). Full hyperparameter configuration per architecture is shown in Table IV.

All experiments were run on an NVIDIA GeForce RTX 3070 Ti GPU (8 GB VRAM) with an AMD Ryzen 5 3600 CPU, using Python 3.10, PyTorch 2.0.1 (CUDA 11.8), torchvision 0.15.2, and scikit-learn. All random seeds were fixed at 42 for reproducibility, with a batch size of 32 for most models and 16 for VGG-16, and 3 DataLoader workers.

Evaluation is performed on the test set (1,919 samples) using Top-1 accuracy and macro F1-score as overall metrics. Macro F1-score is chosen as the primary metric because it assigns equal weight to each class regardless of majority class dominance. As the primary analytical contribution, macro F1-score is also computed separately for the Vowel A group (20 majority classes, 1,518 test samples) and the non-A vowel group (100 minority classes, 401 test samples), with the absolute difference between the two used to measure model generalization balance. As an indicator of statistical robustness, the 95% confidence interval for the macro F1-score is also computed via *bootstrap* on the test-set predictions.

V. RESULTS AND DISCUSSION

A. Overall Performance Comparison

Table V summarizes the performance of all architectures on the test set. EfficientNet-B0 ranks first on every metric with 99.06% accuracy and a macro F1-score of 0.9823 (95% confidence interval 0.9640–0.9929), followed by MobileNetV3-

Large with 98.75% accuracy and a macro F1-score of 0.9663. ResNet-18 ranks third with 97.71% accuracy and a macro F1-score of 0.9321, while VGG-16 closes the ranking with 94.89% accuracy and a macro F1-score of 0.8326. This pattern is notable because the top performers EfficientNet-B0 and MobileNetV3-Large are the lightest architectures with about 5 million parameters, whereas VGG-16 with its 138 million parameters ranks last, so parameter count alone is not deterministic of performance under these imbalanced conditions.

B. Vowel A vs. Non-A Vowel Gap Analysis

The Gap column in Table V reveals a pattern hidden by aggregate metrics, which constitutes one of the primary analytical contributions of this study.

EfficientNet-B0 not only leads on the overall metrics but also maintains the smallest gap of 0.0291, with a Vowel A F1 of 0.9939 against a non-A Vowel F1 of 0.9648. This shows that the structure-preserving augmentation pipeline and the weighted loss function suppress majority-class bias almost entirely, so the model still recognizes minority classes that each have only about 300 augmented training images. MobileNetV3-Large follows with a comparable pattern, namely a gap of 0.0568, consistent with its strength in the study of Wisesty et al. [3] and confirming that a lightweight architecture based on *depthwise separable convolution* remains competitive under the far more demanding 120-class condition.

ResNet-18 displays a qualitatively different pattern. Its gap is indeed small (0.0416), but this stems from a relatively low Vowel A F1 (0.9010) rather than a high non-A Vowel F1, so its balance comes from majority-class performance dropping as well rather than from strengthening of the minority classes. This observation underscores the importance of reading the gap together with both of its underlying values, because a small gap does not always indicate good generalization.

VGG-16 exhibits the largest gap of 0.2154, with its non-A Vowel F1 falling to 0.7232 even though its Vowel A F1 remains high. This reflects a deep convolutional architecture that tends to over-optimize majority classes and is the most sensitive to minority data scarcity.

The per-vowel-group analysis in Table VI deepens this finding by decomposing the macro F1-score for the six vowel groups separately. EfficientNet-B0 leads or ties on almost every group, even reaching a perfect F1 of 1.0000 on the Vowel I group. The most consistent pattern is that the E-taling group is the hardest across all four architectures, with the lowest F1 throughout, ranging from 0.8732 on EfficientNet-B0 down to only 0.6159 on VGG-16. The visual similarity of the E-taling marker to several other vowel markers, together with its scarce sample count, is the likely cause, and this becomes a clear target for improvement in future work.

C. Training Curve Analysis

Fig. 2 and 3 display the training curves of the two best architectures. The Phase 1 to Phase 2 transition (epoch 11) is clearly visible as a sharp loss drop, confirming the effectiveness of the two-phase fine-tuning strategy. EfficientNet-B0 reaches its

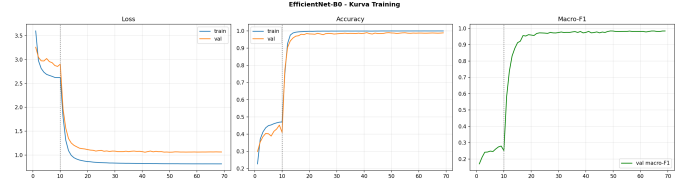


Fig. 2. EfficientNet-B0 training curves. The Phase 1-to-Phase 2 transition is visible at epoch 11 and the best weights are reached at epoch 49.

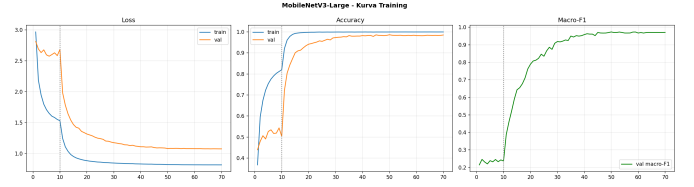


Fig. 3. MobileNetV3-Large training curves as the second-best architecture, with best weights at epoch 50.

best weights at epoch 49 with a validation accuracy of 98.99%, while MobileNetV3-Large reaches its best weights at epoch 50 with a validation accuracy of 98.66%. Both curves show a tight gap between training and validation accuracy, indicating no meaningful overfitting despite the training set being enriched through augmentation. Training curves for ResNet-18 and VGG-16 are omitted as both converged without anomalies, and their epoch counts and full performance metrics are captured in Table V.

D. Prediction Result Visualization

To complement the quantitative analysis, Fig. 4 through 6 visualize the prediction behavior of EfficientNet-B0 as the best architecture on the test set. Fig. 4 presents example predictions on several test images, showing that the model correctly recognizes a range of handwriting styles on most samples, including minority classes. Fig. 5 summarizes the F1-score distribution per vowel group and reaffirms E-taling as the lowest-performing group. Fig. 6 shows a confusion matrix dominated by its diagonal, with the small share of errors concentrated on visually similar character pairs, consistent with the character-similarity findings in prior literature.

E. Comparison with Prior Work

Table VII contextualizes this study within the existing Japanese script recognition literature. This study is the first to evaluate all 120 syllabic classes on a naturally imbalanced dataset, and the first to report four-architecture comparisons with per-vowel-group analysis.

EfficientNet-B0 achieves 99.06% accuracy, surpassing prior balanced studies (97.29% by Susanto et al. [9]) and on par with the highest reported 99.65% on an artificially balanced dataset [8], despite being trained on a dataset with $33.10\times$ imbalance. This confirms that the proposed pipeline successfully bridges naturally imbalanced conditions to match the performance of ideal balanced conditions. Furthermore, no prior study has reported per-vowel-group analysis, making the

TABLE V
PERFORMANCE COMPARISON AND GAP ANALYSIS OF FOUR ARCHITECTURES ON THE 120-CLASS TEST SET

Architecture	Accuracy (%)	Macro F1	Vowel A F1	Non-A F1	Gap	Epochs
EfficientNet-B0	99.06	0.9823	0.9939	0.9648	0.0291	49
MobileNetV3-Large	98.75	0.9663	0.9935	0.9367	0.0568	50
ResNet-18	97.71	0.9321	0.9010	0.8593	0.0416	51
VGG-16	94.89	0.8326	0.9386	0.7232	0.2154	66

$$\text{Gap} = |\text{Vowel A F1} - \text{Non-A F1}|.$$

TABLE VI
MACRO F1-SCORE PER VOWEL GROUP ON THE TEST SET

Vowel Group	Efficient Net-B0	Mobile NetV3-L	ResNet -18	VGG -16
Vowel A	0.9939	0.9935	0.9010	0.9386
Vowel E	0.9889	0.9762	0.8806	0.6383
Vowel I	1.0000	0.9746	0.9214	0.8976
Vowel O	0.9746	0.9378	0.9126	0.5673
Vowel U	0.9873	0.9216	0.8710	0.8969
Vowel E-taling	0.8732	0.8732	0.7111	0.6159

Highest value per row is in bold.

E-taling is consistently the hardest group.

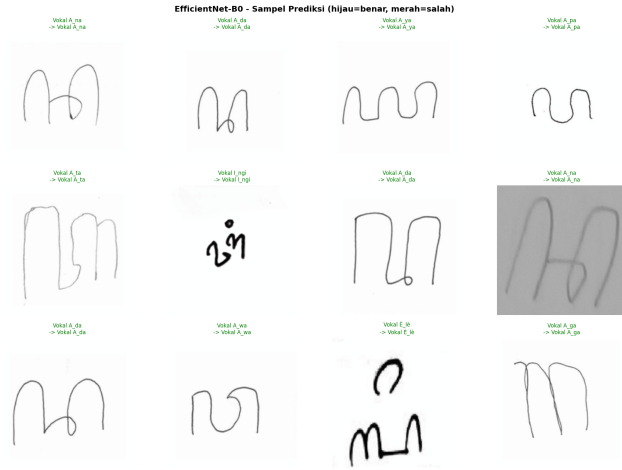


Fig. 4. Example predictions of EfficientNet-B0 on the test set. The true label and predicted label are shown for each sample.

finding that the best generalization balance is achieved by the lightest architecture, and that the E-taling group is consistently the hardest, an entirely novel contribution to this literature.

One limitation must be acknowledged, as each non-A Vowel class is represented by only 4 images in the test set, so a single misprediction already shifts per-class F1 by 25%. Therefore, all conclusions are based on per-vowel-group macro F1 (av-

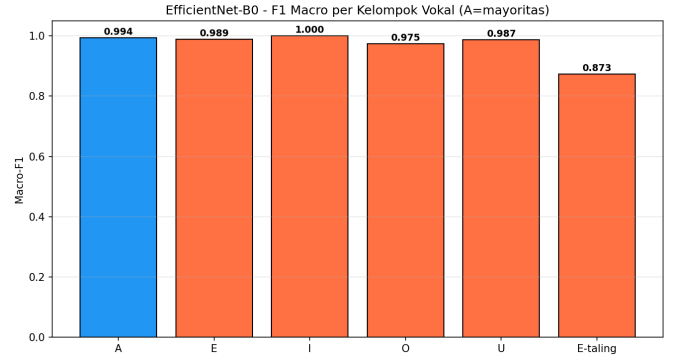


Fig. 5. F1-score per vowel group for EfficientNet-B0. The E-taling group obtains the lowest score.

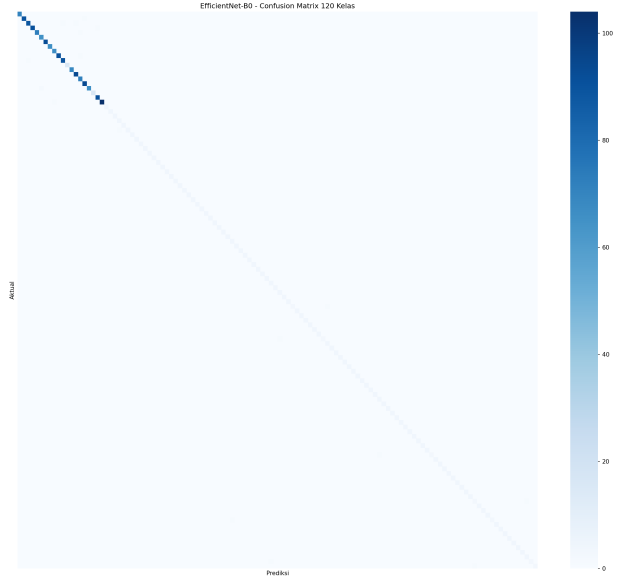


Fig. 6. Confusion matrix of EfficientNet-B0 over 120 classes. The dominant diagonal indicates a low error rate.

eraged over 20 classes), and larger minority data collection remains a priority for follow-up research.

VI. CONCLUSION

This study demonstrates that 120-class handwritten Javanese script recognition under severe natural class imbalance can be

TABLE VII
COMPARISON WITH JAVANESE SCRIPT RECOGNITION STUDIES

Study	Classes	Balanced?	Best Metric
Susanto et al. [9]	120	Yes	Accuracy 97.29%
Susanto et al. [8]	120	Yes	Accuracy 99.65%
Fateah et al. [5]	20	Yes	Accuracy 91%
Wisesty et al. [3]	20	Yes	F1=0.9788
Nindya et al. [6]	20	Yes	Accuracy 93.19%
Faizin et al. [2]	20	No	–
This work	120	No	Accuracy 99.06%

achieved at competitive performance levels through a combination of a structure-preserving data preparation pipeline and transfer learning. EfficientNet-B0 achieves 99.06% accuracy, surpassing prior studies using artificially balanced datasets [9], proving that imbalanced conditions are not an insurmountable obstacle given an appropriate data strategy.

The analytically most significant finding is that the best generalization balance is achieved by the lightest architecture, where EfficientNet-B0 maintains an F1 gap between majority and minority classes of only 0.0291 while remaining the leader on every aggregate metric. The per-vowel-group analysis, never reported in the Javanese script literature before, shows that the E-taling group is consistently the hardest to recognize across all four architectures. This finding has direct practical implications, namely when a Javanese OCR system is deployed on real documents spanning the full vowel spectrum uniformly, EfficientNet-B0 is the most appropriate choice because it combines the highest accuracy with the smallest bias toward majority classes, while dedicated handling of the E-taling group should become a priority for improvement.

This study is limited to isolated character recognition and a small test set for minority classes. Going forward, collecting more minority data will strengthen the validation of these findings, while exploring GAN-based augmentation [2] can complement the proposed photometric approach. Extension to word-level recognition incorporating *pasangan* forms represents the most application-relevant next step [1].

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